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MATERNAL-FETAL MEDICINE

The role of adding metformin in insulin-resistant diabetic pregnant women: a randomized controlled trial

Moustafa Ibrahim Ibrahim · Ahmed Hamdy · Adel Shafik · Salah Taha · Mohammed Anwar · Mohammed Faris

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Abstract
Purpose The aim of the present study is to assess the impact of adding oral metformin to insulin therapy in pregnant women with insulin-resistant diabetes mellitus.
Methods The current non-inferiority randomized controlled trial was conducted at Ain Shams University Maternity Hospital. The study included pregnant women with gestational or pre-existing diabetes mellitus at gestations between 20 and 34 weeks, who showed insulin resistance (defined as poor glycemic control at a daily dose of $\geq$1.12 units/kg). Recruited women were randomized into one of two groups: group I, including women who received oral metformin without increasing the insulin dose; and group II, including women who had their insulin dose increased. The primary outcome was maternal glycemic control. Secondary outcomes included maternal bouts of hypoglycemia, need for another hospital admission for uncontrolled diabetes during pregnancy, gestational age at delivery, mode of delivery, birth weight, birth trauma, congenital anomalies, 1- and 5-min Apgar score, neonatal hypoglycemia, need for neonatal intensive care unit (NICU) admission and adverse neonatal outcomes.
Results A total number of 154 women with diabetes mellitus with pregnancy were approached; of them 90 women were eligible and were randomly allocated and included in the final analysis. The recruited 90 women were randomized into one of two groups: group I (metformin group) ($n = 46$), including women who received oral metformin in addition to the same initial insulin dose; and group II (control group) ($n = 44$), including women who had their insulin dose increased according to the standard protocol. The mean age of included women was 29.84 ± 5.37 years (range 20–42 years). The mean gestational age at recruitment was 28.7 ± 3.71 weeks (range 21–34 weeks). Among the 46 women of group I, 17 (36.9 %) women reached proper glycemic control at a daily metformin dose of 1,500 mg, 18 (39.2 %) at a daily dose of 2,000 mg, while 11 (23.9 %) received metformin at a daily dose of 2,000 mg without reaching proper glycemic control and needed raising the dose of insulin dose.
Conclusion Adding metformin to insulin therapy in women with insulin-resistant diabetes mellitus with pregnancy seems to be effective in proper glycemic control in a considerable proportion of women, along with benefits of reduced hospital stay, reduced frequency of maternal hypoglycemia as well as reduced frequency of neonatal hypoglycemia, NICU admission and neonatal respiratory distress syndrome.

Keywords Diabetes mellitus with pregnancy · Gestational diabetes mellitus · Metformin · Insulin resistance

Abbreviations

AMP	Adenosine monophosphate
ASD	Atrial septal defect
BMI	Body mass index
DM	Diabetes mellitus
FDA	Food and drug administration
GDM	Gestational DM
GIR	Glucose–insulin ratio

M. I. Ibrahim (✉) · A. Hamdy · A. Shafik · S. Taha · M. Faris
Ain-Shams Faculty of Medicine, Abbasiya Square, Cairo, Egypt
e-mail: mi_monem@yahoo.com

M. Anwar
Resident of Gynecology, Gynecological Oncology Unit, Ain Shams University Maternity Hospital, Cairo, Egypt

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HOMA-IR	Homeostatic model assessment of insulin resistance
IQR	Interquartile range
NICU	Neonatal intensive care unit
NNH	Number needed to harm
NNT	Number needed to treat
NPH	Neutral protamine Hagedorn
PCOS	Polycystic ovarian syndrome
PDM	Pre-existing DM
QUICKI	Quantitative insulin sensitivity check index

Introduction

The proportion of pregnant women with gestational and pregestational diabetes mellitus (DM) is increasing, mainly from an increase in type 2 DM [1, 2]. Obesity, low level of physical activity, and, possibly, the exposure to diabetes in utero are major contributors to the increase in type 2 diabetes [3]. Insulin has been the drug of choice for treating DM with pregnancy, because of its safety in pregnancy, lack of significant transplacental passage and long history of use [4]. Pregnant diabetic women do not uncommonly show poor glycemic control that necessitates hospital admission during their antenatal follow-ups. Of them a significant proportion, particularly those who are obese with long-standing pregestational DM, show some insulin resistance, that requires further increase in insulin dose over variable periods of hospital stay. Prolonged hospital stay is known to be associated with several adverse events including a higher risk of nosocomial chest and urinary tract infections, venous thromboembolism, in addition to the high cost. Prolonged hospitalization, on its own, may even worsen glycemic control [5]. Moreover, insulin, in high doses, is associated with weight gain and bouts of hypoglycemia [6]. In the few decades, oral hypoglycemic agents have been included in the armamentarium of treatment modalities for gestational DM (GDM). Earlier concerns with use of these agents in pregnancy were the unknown risk of teratogenicity and neonatal hypoglycemia caused by transplacental passage [4]. Several randomized controlled trials have been conducted on the use of different oral hypoglycemic drugs as a substitute to insulin in women with GDM; the results of all of which were promising. Glycemic control and neonatal outcome in women who received oral hypoglycemic agents, namely, the sulfonylurea glyburide and the biguanide metformin, were comparable to those who received insulin [6]. Metformin is a biguanide that improves insulin sensitivity, probably by activating adenosine monophosphate (AMP) kinase. In contrast to insulin, metformin is not associated with weight gain or hypoglycemia. Reported outcomes of its use during pregnancy have been favorable [7, 8]. The safety of use of metformin during pregnancy was mainly derived from reports of women with polycystic ovarian syndrome (PCOS) who got pregnant while using metformin [4]. Metformin is now categorized by the food and drug administration (FDA) regarding its safety for use in pregnancy as category B [4]. Metformin was tried in women with GDM as an alternative to insulin and showed remarkably promising results [6, 9]. The current recommendation is to switch women with type 2 DM who have been maintained on metformin and get pregnant, to insulin even if unexpected pregnancy occurs [4]. The promising results of the few randomized trials on the use of metformin as an alternative to insulin in women with GDM, encourages us to try it, not as an alternative to insulin, but as an 'adjuvant' in women with DM with pregnancy who show insulin resistance. The aim of the present study was to assess the impact of adding oral metformin to insulin therapy in pregnant women with resistant DM.

Patients and methods

The current non-inferiority randomized controlled trial was conducted at Ain Shams University Maternity Hospital during the period between August 2011 and April 2012. The protocol of the study had been revised and approved by the Ethical Research Committee of the Obstetrics and Gynecology Department, Ain Shams University, and registered online to www.clinicaltrials.gov (ID: NCT01915550). Each patient signed on the informed consent after thorough explanation of the process and purpose of the study, before participating in the study. Each patient had the right to withdraw from the study at any phase, without any adverse impact on her medical or ethical management. The study included pregnant women with gestational or pre-existing DM at gestations between 20 and 34 weeks who showed insulin resistance (defined as poor glycemic control at a daily dose of ≥1.12 units/kg). This threshold level was calculated based on a retrospective survey of randomly selected 100 diabetic pregnant women who had been admitted to Ain Shams University Maternity Hospital over the last 2 years. The range of the total dose of insulin, at which proper glycemic control was reached, was 0.61–1.28 units/kg per day. The threshold level was set at the 90th percentile, which was 1.12 units/kg per day. Poor glycemic control was defined as fasting blood glucose >95 mg/dl and/or 2-h postprandial blood glucose >120 mg/dl [4]. Women with type 1 DM, those with secondary diabetes and those who had liver or renal impairment were not included in the trial. For women who were newly diagnosed to have DM with pregnancy, or those who had not started insulin therapy by the time of admission, insulin was started at a daily dose of 0.7 IU/kg [at

the second trimester of pregnancy (14–27^{+6}weeks' gestation)] or0.8 IU/kg [at the third trimester of pregnancy (≥28 weeks' gestation)]. Insulin dose was increased in women who were admitted for poor glycemic control. Insulin dose was raised at a rate of 1 IU for every 10 mg/dl higher than the target blood glucose concentration. The target blood glucose concentration was 60–95 mg/dl and <120 mg/dl (for fasting and 2-h postprandial status, respectively) [5]. The total dose of insulin was given in two divided doses of a mixture of regular insulin and neutral protamine Hagedorn (NPH) insulin at a ratio of 3:7, 100 IU/ml (Insulin Mixtard®70/30, NovoNordisk, Bagsværd, Denmark): two-thirds in the early morning (7–8 am) and one-third in the evening (5–7 pm). Only women who were admitted to the hospital for poor glycemic control, after reaching a daily dose equivalent to or exceeding the threshold level (1.12 IU/kg), were recruited in the trial. Recruited women were randomized into one of two groups: group I, including women received oral metformin without increasing the insulin dose; and group II, including women who had their insulin dose increased. Randomization was performed using a computer-generated randomization system. To minimize the risk of selection bias, the allocation table was checked after applying eligibility criteria on recruited women. Women who chose to switch from the allocated group to the other one were canceled and not included in the final statistical analysis. Per-protocol treatment analysis was performed. In women of group I, metformin (Cidophage®500 mg, CID Pharmaceuticals, Egypt) was given at a dose of 1,500 mg divided into three doses, were taken with meals, in addition to insulin at the last reached dose. Glycemic response to metformin treatment was assessed by checking fasting and 2-h postprandial blood glucose 5 days after the treatment was started. If the target blood glucose concentrations were not attained yet, the dose of metformin was to be raised to 2,000 mg per day for further 5 days. During these 10 days, daily capillary blood glucose, urine for glycosuria or ketonuria were checked twice daily and fasting and 2-h postprandial blood glucose every 48 h, to detect seriously shooting blood glucose concentrations, and treat it with regular insulin accordingly. Women who did not reach the target blood glucose concentrations 10 days after initiating metformin was switched to the conventional insulin dose-raising regimen. In women of group II, women were kept up with raising insulin dose at the mentioned rate, until reaching the target glycemic control. Again, capillary blood glucose and urine for glycosuria or ketonuria were checked twice daily in addition to fasting and 2-h postprandial blood glucose every 48 h, until reaching the target blood glucose concentrations. Women who showed proper glycemic control in either group were discharged from the hospital and were followed up until delivery. Glycemic control was checked fortnightly until delivery. Women who showed poor glycemic control at any time during pregnancy were admitted and had their insulin dose increased in the previously mentioned pattern until the target blood glucose concentrations were achieved.

The primary outcome was maternal glycemic control. Secondary outcomes included maternal bouts of hypoglycemia (<60 mg/dl), need for another hospital admission(s) for uncontrolled diabetes during pregnancy, gestational age at delivery, mode of delivery, birth weight, birth trauma, congenital anomalies, 1- and 5-min Apgar score, neonatal hypoglycemia, need for neonatal intensive care unit (NICU) admission and adverse neonatal outcomes.

Sample size justification

Sample size was calculated using EpiInfo® version 7.0, setting the power (β) at 80 % and two-sided confidence level at 95 %. The current study was set as non-inferiority randomized controlled trial. All women who have their insulin dose raised would eventually reach a proper glycemic control. A success rate of ≥80 % for metformin to achieve proper glycemic control was assumed to be satisfactory. Calculation according to these values produced a minimal sample size of 43 cases in each group. Assuming a dropout rate of 5 %, a minimal sample size of 45 cases was needed in each group.

Statistical methods

Statistical analysis was performed using Microsoft Excel version 2010 and Statistical Package for Social Sciences (SPSS) for Windows version 15.0. Data were described as range, mean and standard deviation (for numeric parametric variables); range, median and interquartile range (IQR) (for numeric non-parametric variables), or number and percentage (for categorical variables). Difference between two independent groups was estimated using independent student's t test (for numeric parametric variables), Mann–Whitney's U test (for numeric non-parametric variables), or χ^2 (for categorical variables). Yates' continuity correction was applied to χ^2 test if one or more of the expected values were less than 5. Significance level was set at 0.05.

Results

A total number of 154 women with DM with pregnancy were approached; of them 90 women were eligible and were randomly allocated and included in the final analysis (Fig. 1). The mean age of included women was 29.84 ± 5.37 years (range 20–42 years). The median

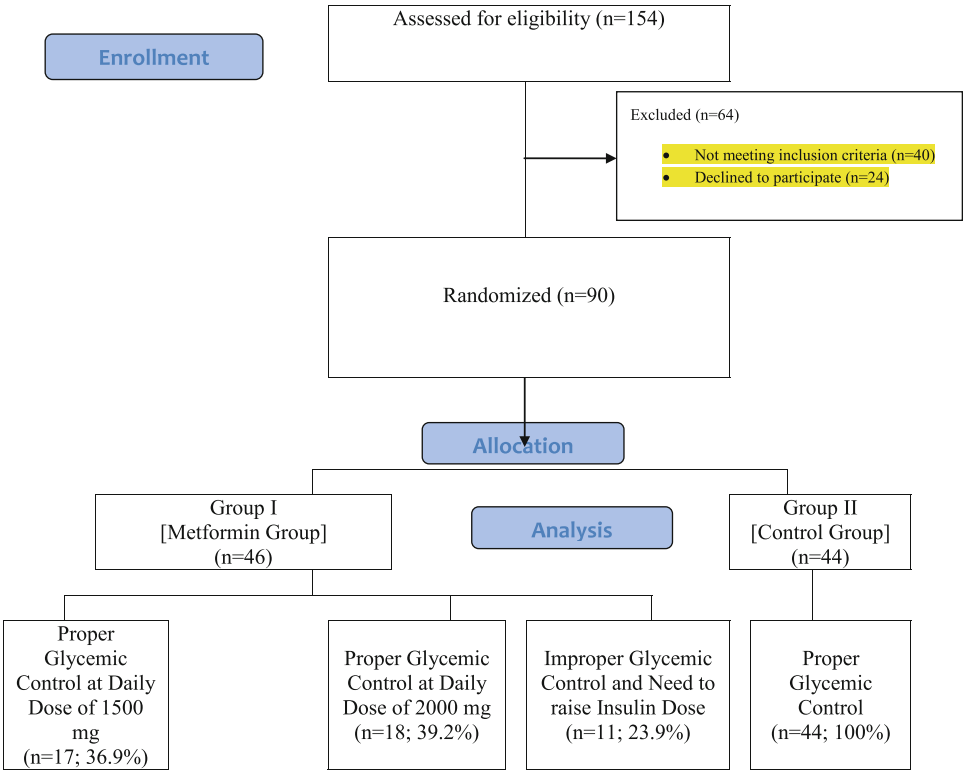

Fig. 1 CONSORT 2010 flow diagram

parity was 2 (range 0–6; IQR 2–3). The median number of previous miscarriages was 0 (range 0–6; IQR 0–1). The mean gestational age at recruitment was 28.7 ± 3.71 weeks (range 21–34 weeks). The mean body mass index (BMI) was 31.83 ± 3.23 kg/m^2 (range 25.71–43.37 kg/m^2). Of the included 90 women, 64 (71.1 %) were obese (BMI ≥30 kg/m^2), while 26 (28.9 %) were non-obese; 39 (43.3 %) had GDM, while 51 (56.7 %) had pre-existing DM (PDM). The median disease duration in women with PDM was 4 years (range 1–15 years; IQR 3–5.5 years). The median initial total insulin dose was 1.14 IU/kg per day (range 1.12–1.3 IU/kg per day; IQR 1.13–1.17 IU/kg per day). The recruited 90 women were randomized into one of two groups: group I (metformin group) ($n = 46$): including women who received oral metformin in addition to the same initial insulin dose; and group II (control group) ($n = 44$): including women who had their insulin dose increased according to the standard protocol. There were no significant differences between women of both groups regarding age, parity, no. of previous miscarriages, gestational age at recruitment and BMI. There were no significant differences between women of both groups regarding the type and duration of DM.

Difference between groups regarding glycemic control is shown in Table 1. Of the included 90 women, 8 (3 in group I and 5 in group II) were inaccessible for follow-up at delivery (due to loss of contact), comprising a dropout rate of 8.9 %. Table 2 shows difference between both groups regarding delivery outcomes. There was a significant association between obesity and failed glycemic control among women of group I (Table 3).

Discussion

The current trial showed that adding metformin, as an 'outpatient adjuvant' treatment to insulin in women with DM with pregnancy (whether gestational or pre-existing) who showed insulin resistance, was associated with proper glycemic control in 76.1 % of treated women (NNH = 4; i.e., in every 4 women who showed insulin resistance and received metformin, 3 reached glycemic control and 1 needed to raise the insulin dose). Although this reduction in the rates of proper glycemic control was statistically significant; it was not, from the clinical point of view. Adding metformin, rather than increasing the insulin dose, was associated with a significant reduction in the rates of hospitalization, a significant reduction in the treatment cost (the average overall added cost was 60–80 EGP with metformin, in contrast to 450–950 EGP with raising the insulin dose), as well as a significant reduction in the frequency of maternal hypoglycemia (NNT = 7). Moreover, adding metformin, rather than raising the insulin dose, was associated with a significant reduction in the rates of neonatal hypoglycemia (NNT = 3) and NICU admission (NNT = 4) and a modest reduction in the rates of neonatal

Table 1 Difference between groups regarding glycemic control

	Group I (metformin group) ($n = 46$)	Group II (control group) ($n = 44$)	*P*	ARR	NNT/NNH
Proper glycemic control	35 (76.1 %)	44 (100 %)	0.001	−23.9 %	NNH = 4
Readmission for poor glycemic control	6 (13 %)	8 (18.2 %)	0.471	5.1 %	NNT = 19
Bouts of maternal hypoglycemia	3 (6.5 %)	10 (22.7 %)	0.029	16.2 %	NNT = 6

Data presented as number (percentage)

ARR Absolute risk reduction, *NNT* number needed to treat, *NNH* number needed to harm

Table 2 Difference between groups regarding delivery outcomes

	Group I (metformin group) ($n = 43$)	Group II (control group) ($n = 39$)	*P*	ARR	NNT/NNH
Mode of delivery					
Vaginal delivery	13 (30.2 %)	12 (30.8 %)	0.958	−0.6 %	NNH = 186
Cesarean section (CS)	30 (69.8 %)	27 (69.2 %)			
Indication for cesarean section (CS)					
Failed progress of labor	8 (18.6 %)	5 (12.8 %)	0.474	5.8 %	NNT = 17
Fetal macrosomia	7 (16.3 %)	6 (15.4 %)	0.912	−0.9 %	NNH = 112
Fetal compromise	5 (11.6 %)	7 (17.9 %)	0.419	6.3 %	NNT = 16
Other indications[a]	10 (23.3 %)	9 (23.1 %)	0.985		
Gestational age at delivery (weeks)	37.89 ± 0.32	37.7 ± 0.66	0.250	–	–
Birth weight (g)	3,757.89 ± 298.26	3,790 ± 403.47	0.780	–	–
Fetal macrosomia	10 (23.3 %)	12 (30.8 %)	0.443	7.5 %	NNT = 13
1-min Apgar Score	7 (5–8)	6 (5–7)	0.180	–	–
5-min Apgar Score	8 (8–9)	8 (8–9)	0.693	–	–
Congenital malformations	1 (2.3 %) (one case of cleft lip)	1 (2.6 %) (one case of ASD)	0.518	0.24 %	NNT = 419
Neonatal hypoglycemia	3 (7 %)	15 (38.5 %)	0.001	31.5 %	NNT = 3
NICU admission	8 (18.6 %)	16 (41 %)	0.026	22.4 %	NNT = 4
Stillbirth	0 (0 %)	2 (5.1 %)	0.431	5.1 %	NNT = 20
RDS	5 (11.6 %)	10 (25.6 %)	0.176	14 %	NNT = 7

Data presented as number (percentage), mean ± SD or median (interquartile range)

NICU Neonatal Intensive Care Unit, *RDS* respiratory distress syndrome, *ASD* atrial septal defect, *ARR* absolute risk reduction, *NNT* number needed to treat, *NNH* number needed to harm

[a] Other indications included previous CS, placenta previa or malpresentation

RDS (NNT = 7). The presence of obesity was significantly associated with reduced rates of proper glycemic control with metformin.

There has been emerging evidence regarding the safety of the use of metformin in pregnancy; most of it was derived from studies related to the use of metformin in women complaining of infertility secondary to PCOS [10–12]. PCOS is known to be closely related to insulin resistance, even if overt DM is not formally diagnosed. It has been shown that metformin does reduce insulin resistance and, therefore, improves outcomes in women with PCOS [13]. Meanwhile, over the past decade, there has been strong evidence regarding the efficacy and the safety of use of metformin in women with GDM, as an alternative to the standard insulin therapy [6, 9, 14–19].

Ijas et al. [20] in their open-label prospective randomized controlled study reported that metformin seems to be a relevant drug for the prevention of fetal macrosomia, mainly in lean or moderately overweight women developing GDM in late pregnancy. Women with ample obesity, high fasting blood glucose and an early demand for pharmacological therapy may be more suitable for insulin treatment.

The results from our study are consistent with those obtained by Niromanesh et al. [21], who were assigned randomly 180 women to receive either metformin ($n = 80$) or insulin ($n = 80$). They assessed the effect of metformin and insulin in glycemic control and compared the pregnancy outcome in women with GDM and they authenticated that, metformin is an effective and safe alternative

Table 3 Association between among women of group I

Group I	Proper glycemic control	Failed glycemic control	*P*	RR (95 % CI)
Obesity				
Obese (BMI ≥30 kg/m^2)	20 (57.1 %)	11 (100 %)	0.023	0.6
Non-obese (BMI < 30 kg/m^2)	15 (42.9 %)	0 (0 %)		(0.43–0.76)

Data presented as number (percentage)

BMI body mass index, *RR* (95 %) relative risk CI (95 %) confidence interval

treatment to insulin for women with GDM. Their study does not display significant risk of maternal or neonatal adverse outcome with the use of metformin.

Similar to our findings, a more recent study was conducted on 28 women with pre-existing type 2 diabetes and those diagnosed with GDM who required medical treatment prior to 20 weeks. Those women who were randomly assigned to metformin ($n = 14$) and insulin ($n = 14$), their glucose measures did not differ between the groups; and the percentage of women who met fasting and postprandial glycemic target figures did not differ between the groups. Women treated with metformin had significantly fewer subjective attacks of hypoglycemia compared with those using insulin (0 versus 36 %; $p = 0.04$), as well as documented glucose values <60 mg/dL (7.1 versus 50 %; $p = 0.03$). From the previously mentioned results, Hickman et al. [22] concluded that metformin should be considered for the treatment of overt diabetes and early A2 GDM in pregnancy.

A beneficial role of adding metformin in women who show insulin resistance was suggested by Simmons et al. [23]. The strength of the current study included that it is the first conducted randomized controlled trial that investigated the efficacy of adding metformin to insulin therapy in women with DM with pregnancy that shows insulin resistance.

The definition of insulin resistance remains controversial. Several indices based on serum insulin and glucose levels were proposed, including glucose–insulin ratio (GIR), homeostatic model assessment of insulin resistance (HOMA-IR) and quantitative insulin sensitivity check index (QUICKI) [24]. In the current trial, none of these indices were used to define insulin resistance, however. No reference ranges for such indices were previously published for pregnant women. Moreover, we preferred to select a clinical marker of insulin resistance that we face in the everyday clinical work in our institute, namely, a high-insulin-dose requirement. To be quite objective, retrospective analysis was performed over a randomly selected 100 cases of uncontrolled DM with pregnancy that had been admitted to the hospital in the preceding 6 months. Levels above the 90th percentile of the daily insulin doses at which glycemic control was achieved were considered as 'high insulin doses' that probably reflect insulin resistance (the first limitation of the current study). Another limitation of the current study was the estimation of the sample size based on a non-inferiority assumption. To conclude a more valid generalizable result, further studies with larger sample size (that should be estimated on a superiority assumption, instead) are probably needed. A third limitation was probably the 'heterogeneity' in the BMI of the included women (range 25.71–43.37 kg/m^2). Both insulin resistance and efficacy of metformin are known to be affected by obesity, a fact that was even confirmed by the results of the current trial. Future trials should consider this limitation and should clearly include a more homogeneous group of women regarding their BMI.

Conclusion

Adding metformin to insulin therapy in women with insulin-resistant DM with pregnancy seems to be effective in proper glycemic control in a considerable proportion of women, with reduced hospital stay, reduced frequency of maternal hypoglycemia as well as reduced frequency of neonatal hypoglycemia, NICU admission and neonatal RDS.

Conflict of interest The authors reported no conflict of interest. All the authors had substantial contributions to conception and design, acquisition of data, analysis and interpretation of data, drafting and revising the article critically with final approval of the version to be published. The research was funded by the authors.

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